

Impact of Clinical Pharmacist-led Interventions on the Outcomes of Patients with Bipolar I Disorder: A Randomized Clinical Trial

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ABSTRACT

Objective: Poor medication adherence, drug interactions, and adverse drug events occur frequently in patients with bipolar I disorder (BD-I), affecting their treatment outcomes. Due to limited research regarding the impact of pharmaceutical care (PC) services in the management of patients with BD-I, this study was designed to assess the role of clinical pharmacist-led interventions on outcomes of BD-I patients. **Methods:** A prospective randomized clinical trial was designed, and 59 patients were randomly assigned to the intervention group and 48 patients to the control group. Patients in the intervention group were provided with medication therapy management and follow-up services by the clinical pharmacist, whereas the control group only received routine care. Outcomes which were assessed at baseline (before discharge), 1 month, and 3 months after discharge were the Medication Appropriateness Index (MAI), Beck Depression Inventory-II (BDI-II), Young Mania Rating Scale, and World Health Organization Quality of Life, Brief version (WHOQOL-BREF). **Findings:** Endpoint mean changes in MAI scores from baseline were -5.25 ± 5.19 and 2.02 ± 3.98 points for the intervention and control groups, respectively ($P < 0.001$). Depressive symptoms, measured by the BDI-II, also showed significant improvement in the intervention group; the mean change from baseline to 2nd follow-up assessment was -1.47 ± 7.73 in the intervention group and 1.66 ± 6.42 in the control group ($P = 0.02$). Furthermore, the mean change from baseline to 2nd follow-up in the psychological health domain of the WHOQOL-BREF questionnaire was significantly higher in the intervention group (4.59 ± 17.79) compared with the control group (-3.90 ± 12.55) ($P = 0.005$). **Conclusion:** Our findings reveal that clinical pharmacist-provided services could positively affect outcomes in BD-I patients.

KEYWORDS: Bipolar disorder; clinical pharmacist-led care; Medication Appropriateness Index; pharmaceutical services

INTRODUCTION

Over the previous years, the responsibilities of pharmacists have shifted dramatically from their conventional drug-focused role to a more patient-centered approach.^[1] The term “pharmaceutical care” (PC) was first introduced in 1975.^[2] A contemporary conceptualization proposed by the PC Network Europe characterizes PC as a professional service exclusively provided by pharmacists. This definition presents PC as the pharmacists’ specialized contribution to patient-centered care, with the primary objective of optimizing

medication use and health outcomes.^[3] This practice contains a broad range of services such as medication therapy management, consultations on medication adherence, lifestyle modifications, and health-related education that lead to improved treatment effectiveness and patient safety.^[4]

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Bipolar disorder (BD) is characterized as a severe and debilitating psychiatric condition that causes substantial functional impairment^[5] and decreased Quality of Life (QoL) in affected individuals, even during euthymic states.^[6,7] A comprehensive systematic review analyzing cost-of-illness studies for BD estimated that the worldwide annual economic burden associated with this condition ranged from US \$1904 to \$33,090 per person in 2013.^[8] Furthermore, a recent study in 2022 demonstrated that the annual cost of BD was \$4227 per patient in Iran.^[9]

Pharmacological interventions constitute the foundation of successful treatment for all individuals diagnosed with BD.^[10] Issues such as poor medication adherence,^[11] potential drug interactions, and adverse drug events are integral components of a comprehensive BD treatment plan. One of the most frequently used medications for the treatment of this disorder, i.e. lithium, has a narrow therapeutic index with numerous drug–drug and food–drug interactions, necessitating close monitoring and patient education.^[12]

Considering the role of pharmacists in improving outcomes of patients with severe and persistent mental disorders,^[13] and given the limited research regarding the effectiveness of clinical pharmacist interventions in patients with bipolar I disorder (BD-I), along with the need for more evidence derived from randomized clinical trials, the present study was designed to assess whether PC services provided by a clinical pharmacist, as a member of a multidisciplinary team, could improve the outcomes of patients with BD-I.

METHODS

The current research was a prospective randomized clinical trial on patients with BD-I. This study was carried out in two adult acute care wards and adult outpatient psychiatric department in a tertiary care hospital from June 2022 to September 2023. The study procedure was approved by the institutional ethics committee and was registered on the Iranian Registry of Clinical Trials (registration No: IRCT20210307050613N1).

Few days before discharge, patients were screened for enrollment eligibility and were recruited if they could fulfill the following criteria: having a *Diagnostic and Statistical Manual of Mental Disorders*, Fifth Edition diagnosis of BD-I,^[14] aged 18 years and older, and receiving at least one mood stabilizer or antipsychotic. The exclusion criteria are mentioned in Figure 1.

After verification of the above criteria and obtaining patients' informed consent to participate in the study, they were randomly assigned through simple randomization

to control or intervention groups to receive either usual care or a combination of usual care and PC services, respectively. The sample size of this study was 112 participants, and the patients were followed up for 3 months after discharge. The flow diagram of patient recruitment, allocation, randomization, and analysis is demonstrated in Figure 1.

A comprehensive monitoring form was utilized to record all clinical and demographic information of the patients. In both groups, demographic data, patient's medical and medication history, and relevant clinical information were obtained through a comprehensive review of medical records. For the intervention group, additional clinical information required for pharmaceutical interventions was gathered through interviewing with patients and, where necessary, with their caregivers. Patients in the control group received usual care, which included scheduled visits by physician, general instructions for medication administration at the time of discharge, and consultation with other medical services on demand, whereas in the intervention group, participants were served with multifaceted clinical pharmacist-led services in addition to the usual care. For patients in the control group, the pharmacotherapy intervention was discussed with the responsible physician solely in potentially life-threatening cases.

The PC interventions were as follows: (1) comprehensive medication review followed by recommendations for healthcare professionals or patients before discharge as well as during follow-up visits; (2) verbal education focused on medications and disease state information before discharge and briefly at each outpatient visit; comprehensive education about medication(s) in cases of modifications to the drug regimen at outpatient visits; and (3) an individualized approach to the reason(s) for medication nonadherence and providing verbal education to enhance medication compliance before discharge and briefly at each outpatient visit; patients with a history of acceptable adherence or inadvertent nonadherence received general education concerning medication compliance. All of the specified domains of interventions were initiated after mood stabilization and within the last few days of hospitalization. Educational materials were carefully tailored to match the patients' comprehension levels.

In both groups, comprehensive outcome assessments were conducted at three time points including a few days before discharge, as well as 1-month and 3-month postdischarge, using standardized questionnaires and scoring tools. At each of the specified time points, patients were asked to complete self-administered questionnaires. Data required for the clinician-rated questionnaires were

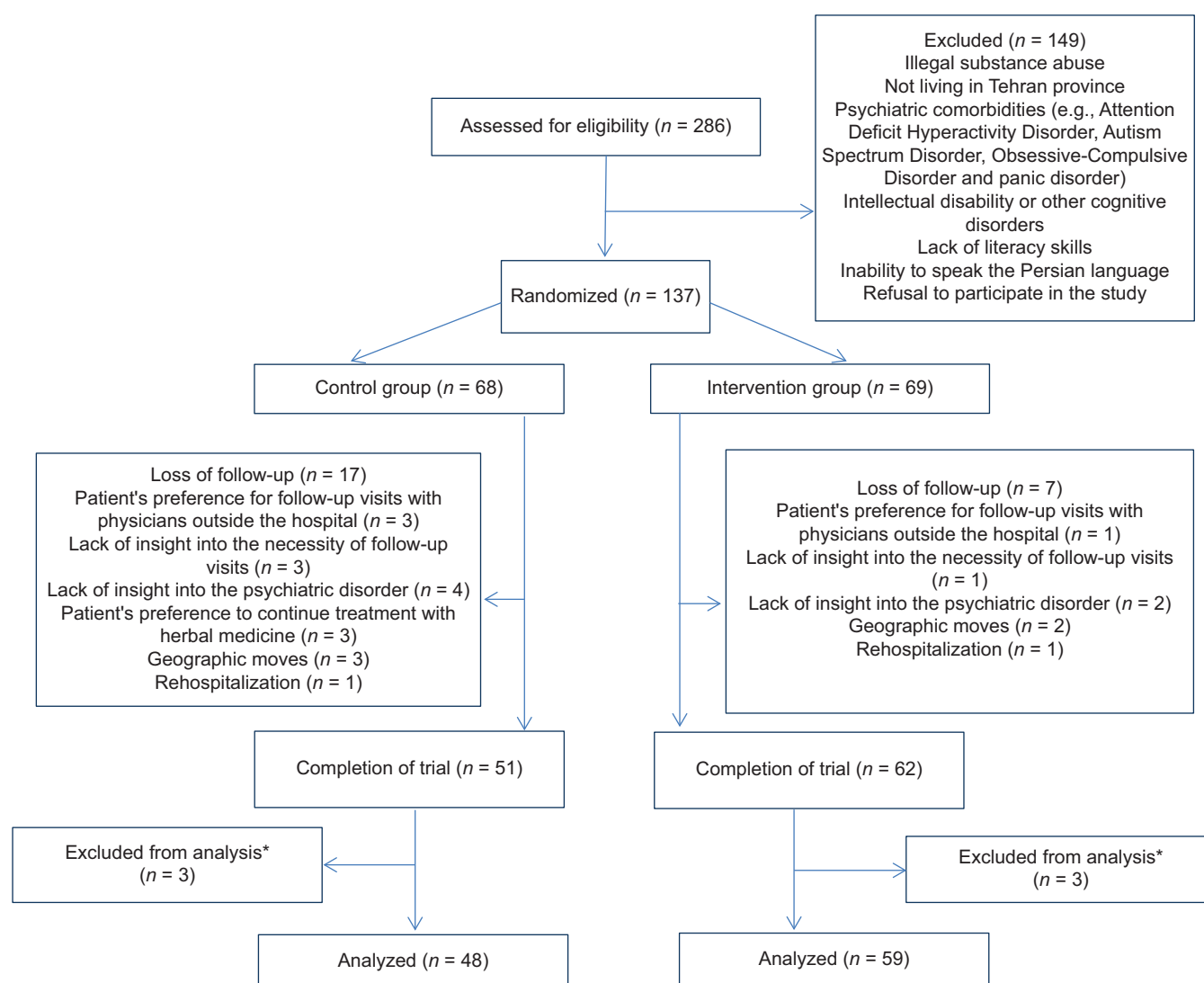


Figure 1: Flow diagram of patients during the study. *Male patients were excluded from the analysis owing to their low population (predominantly not included in the study due to addiction)

obtained through medical records and patient interviews. All of the required information was collected in a state of alertness and free from somnolence. To evaluate prescribing quality and medication suitability as the primary outcome, the Medication Appropriateness Index (MAI), which is a weighted scoring system comprising ten clinician-rated criteria, was used. MAI is calculated for each medication in the drug regimen with respect to its indication, effectiveness, dosage, usage directions, applicability, drug–drug interactions, drug–disease interactions, duplication, duration of therapy, and expense; a higher MAI score indicates a more severe condition.^[15] Furthermore, we used the Persian version of Beck Depression Inventory-II (BDI-II) which is a 21-item self-report questionnaire to evaluate depressive symptoms during the preceding 2 weeks.^[16,17] For the assessment of manic symptoms in patients, the 11-item clinician-rated Young Mania Rating Scale (YMRS) was

implemented.^[18] For the assessment of QoL, the World Health Organization QoL, Brief version (WHOQOL-BREF) questionnaire was used; this self-report, multidimensional measure evaluates four domains of QoL including physical health, psychological health, social relationships, and environmental health with 26 questions.^[19,20]

Before each follow-up visit, a phone call was made to patients in both groups to remind them of their appointments. The total time allocated for each patient in the intervention group was approximately 240 min throughout the study.

Qualitative variables were expressed as frequencies and percentages. Differences between the intervention and control groups with respect to qualitative variables were analyzed using the Chi-square test or the Fisher's exact test, where appropriate.

Since the sample size in both groups was >30, the sampling distribution was considered normal for statistical analysis according to the central limit theorem;^[21] therefore, the appropriate parametric statistical tests were applied for the comparison of the quantitative data between study groups.

Quantitative data were demonstrated as mean and standard deviation. Independent samples *t*-test was used to analyze differences between groups with respect to mean and mean changes. All statistical analyses were performed using IBM SPSS Statistics 27.0.1 and $P < 0.05$ was considered statistically significant.

RESULTS

A total of 286 patients were screened, and 137 were enrolled in the study; of these, 69 patients were assigned to the intervention group and 68 to the control group. Throughout the study, 17 patients in the control group and seven patients in the intervention group dropped out. Finally, out of 113 patients who completed the follow-up visits, results for 48 patients in the control group and 59 patients in the intervention group were analyzed; to enhance the homogeneity of the study patients, data related to the male participants were excluded from analysis ($n = 6$) due to their low population [Figure 1].

As demonstrated in Table 1, there were no statistically significant differences between the control and intervention groups with respect to baseline characteristics. Furthermore, Table 2 shows that no statistically significant differences were detected regarding baseline outcome variables in between-group analyses. The dominant physical comorbidities in both groups were iron deficiency anemia, hypothyroidism, and dyslipidemia. Among the participating patients, the most frequent psychiatric treatment was a combination of an atypical antipsychotic and a mood stabilizer.

At baseline, no significant difference was observed between the groups regarding medication appropriateness as measured by MAI. Throughout the study, the intervention group showed a substantial decrease in MAI scores compared with the baseline status, while the control group's MAI scores increased; consequently, both the 1st and 2nd follow-up assessments revealed statistically significant differences between the intervention and control groups ($P < 0.001$). According to Table 3, regarding MAI, the endpoint mean changes from baseline were -5.25 ± 5.19 points for the intervention and 2.02 ± 3.98 points for the control group ($P < 0.001$).

In this study, depressive symptoms as measured by BDI-II were significantly improved in the intervention

Table 1: Baseline demographic characteristics of patients in the intervention and control groups

Variables	Control group (n=48)	Intervention group (n=59)	P
Age (year)	35.87±12.35	37.25±10.81	0.54*
Duration of illness (year)	9.52±8.62	7.50±7.04	0.18*
Number of prior hospitalizations	2.10±2.94	1.33±1.64	0.11*
Cause of recent hospitalization			
Manic episode	7 (14.58)	9 (15.25)	0.20†
Manic episode + PF	6 (12.5)	19 (32.2)	
Manic episode + MF	13 (27.08)	13 (22.03)	
Manic episode + MF + PF	7 (14.58)	5 (8.47)	
Depressive episode	9 (18.75)	9 (15.25)	
Depressive episode + PF	4 (8.33)	3 (5.08)	
Depressive episode + MF	0	1 (1.69)	
Depressive episode + MF + PF	2 (4.16)	0	
Education level			0.31‡
High school or less	14 (29.16)	12 (20.33)	
Diploma	25 (52.08)	29 (49.15)	
Academic degree	9 (18.75)	18 (30.50)	
Occupational status			0.18‡
Employed	17 (35.41)	14 (23.72)	
Unemployed	31 (64.58)	45 (76.27)	
Marital status			0.46‡
Single	27 (56.25)	44 (74.57)	
Married	21 (43.75)	15 (25.42)	
Smoking status			0.59‡
Smoker	22 (45.83)	24 (40.67)	
Nonsmoker	26 (54.16)	35 (59.32)	

*Independent samples *t*-test, †Fisher's exact test, ‡Chi-square test. Data are presented as mean±SD or *n* (%). MF=Mixed feature, PF=Psychotic feature, SD=Standard deviation

group in comparison with the control group at 1st and 2nd follow-up assessments ($P = 0.04$ and 0.001 , respectively); mean changes from baseline to 2nd follow-up were -1.47 ± 7.73 points for the intervention group and 1.66 ± 6.42 for the control group ($P = 0.02$). Nevertheless, with respect to manic symptoms, as determined by YMRS, no statistically significant intergroup differences were observed across any of the follow-up evaluations.

As shown in Table 2, comparison of the intervention and control groups regarding psychological, social, and environmental health domains of WHOQOL-BREF, revealed significant differences between groups at the 2nd follow-up evaluations. For the psychological health domain, the difference was significant at the 1st follow-up as well. However, analysis of mean changes from baseline to 2nd follow-up revealed comparable outcomes between the groups across most domains, with the notable exception of the psychological health domain, where a significant between-group difference was observed ($P = 0.005$).

Table 2: Between-group analysis of outcomes at baseline and follow-up visits

Variables	Control group (n=48), mean±SD	Intervention group (n=59), mean±SD	P*	95% CI of the difference	
				Lower bound	Upper bound
MAI					
Baseline visit	7.02±5.08	7.45±5.43	0.67	-2.47	1.59
Follow-up visit 1	9.16±5.87	4.49±4.88	<0.001	2.61	6.73
Follow-up visit 2	9.04±5.71	2.20±2.73	<0.001	5.04	8.63
YMRS					
Baseline visit	1.89±2.16	2.59±2.34	0.11	-1.57	0.17
Follow-up visit 1	1.02±1.49	1.11±1.48	0.73	-0.67	0.47
Follow-up visit 2	1±1.87	0.84±1.81	0.67	-0.55	0.86
BDI-II					
Baseline visit	9.95±6.31	8.84±6.86	0.39	-1.44	3.66
Follow-up visit 1	11.22±8.41	8.35±6.06	0.04	0.09	5.65
Follow-up visit 2	11.62±7.58	7.37±5.73	0.001	1.69	6.80
QoL-physical health					
Baseline visit	59.59±17.26	65.55±13.68	0.055	-12.04	0.12
Follow-up visit 1	57.51±14.44	61.98±10.33	0.07	-9.40	0.46
Follow-up visit 2	56.32±16.06	62.04±14.79	0.058	-11.64	0.25
QoL-psychological health					
Baseline visit	48.35±16.41	53.60±16.56	0.10	-11.60	1.10
Follow-up visit 1	48.78±14.78	55.79±14.37	0.01	-12.61	-1.39
Follow-up visit 2	44.44±18.54	58.19±14.28	<0.001	-20.22	-7.26
QoL-social relationships					
Baseline visit	47.74±26.56	56.77±23.54	0.06	-18.64	0.57
Follow-up visit 1	46±21.87	51.27±19.50	0.19	-13.20	2.67
Follow-up visit 2	43.22±22.45	52.96±17.69	0.01	-17.43	-2.04
QoL-environmental health					
Baseline visit	60.61±13.88	65.99±14.70	0.056	-10.91	0.14
Follow-up visit 1	61.06±15	66.10±12.93	0.06	-10.39	0.32
Follow-up visit 2	62.10±13.99	68.27±12.30	0.01	-11.20	-1.11

*Independent samples *t*-test. SD=Standard deviation, MAI=Medication appropriateness index, YMRS=Young Mania Rating Scale, BDI-II=Beck Depression Inventory II, QoL=Quality of life, CI=Confidence interval

Table 3: Comparison of mean changes from baseline to 2nd follow-up visit in study outcomes between the intervention and control groups

Variables	Control group (n=48), mean±SD	Intervention group (n=59), mean±SD	P*	95% CI of the difference	
				Lower bound	Upper bound
MAI	2.02±3.98	-5.25±5.19	<0.001	5.51	9.03
YMRS	-0.89±2.83	-1.74±2.24	0.08	-0.12	1.82
BDI-II	1.66±6.42	-1.47±7.73	0.02	0.37	5.90
QoL-physical health	-3.27±12.43	-3.51±15.64	0.93	-5.27	5.74
QoL-psychological health	-3.90±12.55	4.59±17.79	0.005	-14.32	-2.66
QoL-social relationships	-4.51±13.19	-3.81±15.26	0.80	-6.24	4.84
QoL-environmental health	1.49±9.36	2.27±11.39	0.70	-4.84	3.28

*Independent samples *t*-test. SD=Standard deviation, MAI=Medication appropriateness index, YMRS=Young Mania Rating Scale, BDI-II=Beck Depression Inventory II, QoL=Quality of life, CI=Confidence interval

DISCUSSION

The results of the current study suggest encouraging outcomes from the intensive clinical pharmacist-led services.

Potentially inappropriate prescribing is a major health concern and in this research, the extent of prescription appropriateness was determined using MAI, which

demonstrated a significant disparity between the study groups. The improvement in MAI scores of the intervention group resulted from better scores in the criteria including indication, dosage, applicability, and duplication. This finding is consistent with the results of the study in patients with different psychiatric disorders conducted by Wolf and colleagues, in which pharmacist interventions were associated with significantly improved

MAI scores.^[22] It has been shown that MAI could be considered as a predictor of health-related QoL,^[23] and higher MAI scores are related to unscheduled outpatient or emergency department visits.^[24]

BD is a chronic and disabling psychiatric condition, characterized by alternating episodes of mania or hypomania and depression, causing significant fluctuations in mood, energy, and cognitive function.^[14] In this research, patients in both groups experienced improvement in manic symptoms throughout the study and pharmaceutical interventions did not lead to significant between-group differences in this study outcome. However, in the intervention group, depressive symptoms improved significantly compared with those of the control group. These findings are consistent with the study by Salazar-Ospina *et al.*, in which, despite the absence of significant changes in mania scores compared with the control group, a statistically significant improvement was observed in depression severity as measured by the Hamilton Depression Rating Scale.^[25] In another study, the implementation of pharmacist interventions for individuals diagnosed with major depressive disorder was associated with a significant reduction in BDI-II scores;^[26] one explanation for the absence of significant changes in manic symptoms is the fact that subsyndromal depressive symptoms occur approximately three times more frequently than manic symptoms in individuals with BD;^[27] furthermore, lithium, the most frequently prescribed mood stabilizer in our study, exhibits differential onset of action for manic and depressive symptoms; while manic symptom resolution typically occurs within 1–3 weeks of lithium initiation, the full therapeutic effect on depressive symptoms may require up to 2 months to be manifested;^[28] consequently, patients in both groups were more likely to experience persistent depressive symptoms throughout the study period, rather than manic presentations. The improvement in MAI criteria scores, including indication, dosage, and duplication, led to more appropriate treatment plans and discontinuation of medications without a clear indication, as well as initiating medications for the treatment of comorbidities such as anemia that have been left untreated previously, is also effective in improving depressive symptoms in the intervention group. Resolution of depressive symptoms is highly important in bipolar patients; in comparison with manic and hypomanic episodes, depressive presentations are associated with more severe impairment in functional capacity.^[29] Therapeutic interventions specifically aimed at mitigating depressive symptoms may potentially attenuate the economic burden of this disorder.^[30] In an internet-based survey conducted across 11 countries, >40% of bipolar patients

identified the better treatment of depressive symptoms as a major priority.^[31]

Patients diagnosed with BD typically report a diminished QoL and impaired functional capacity relative to the general population.^[32] The comparison of the intervention and control groups regarding psychological, social relationships, and environmental health domains of WHOQOL-BREF, revealed significant between-group differences at the 2nd follow-up visits; this finding is consistent with two previous studies conducted by Mishra *et al.*^[33] and Singh *et al.*^[34] that indicated significant improvement in different domains of QoL following pharmacist interventions in bipolar patients. However, only the psychological health domain exhibited a significant mean change between the two groups over the course of our study; this might be attributed to the study's focus on recently hospitalized patients with a higher disease burden, and intervention effects on QoL improvement might be less noticeable.

The attrition rate for the control group was approximately 25%, which was more than twice that of the intervention group; this difference indicates that intervention patients likely had higher satisfaction, better illness insights, and decreased number of missed follow-up visits after discharge.

In conclusion, we found that clinical pharmacist interventions have the potential to improve the outcomes of patients living with BD-I; patients in the intervention group experienced lower depressive symptoms, had better MAI scores as well as better QoL in the psychological health domain.

A limitation to our study is the relatively short postdischarge follow-up period, potentially restricting our ability to evaluate the long-term effects of the pharmacist-led services. The unblinded nature of assessments is another limitation that could serve as a source of bias.

For future research, conducting multicenter studies on male and female BD-I patients with the contribution of several pharmacists is suggested, which could result in a better approximation of the real world. Finally, evaluating the effectiveness and cost of low-intensity pharmaceutical interventions in BD-I patients, especially for resource-limited settings, can help identify the minimum interventions needed to improve outcomes and provide more cost-effective strategies for pharmaceutical interventions in this population.

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Conflicts of interest

There are no conflicts of interest.

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